

Building the strand-1 agent, a design note

Revised reading, 2026-09-22 (Jonas's decision 3: update the documents) This note's specification of strand 1 (§1 to §4) was checked against Engström et al. (2024) and holds. Its card ladder (§5) is superseded: its S1.4 (free following) was withdrawn on Jonas's objection that a car-following model would be what got measured; its S1.5 (equilibrium) falls to the same objection; the cards actually run under these numbers are different cards (out/s14_strand1_preference.md, out/s15_comfort_threshold.md, out/s16_admissibility.md), all not credited. Two of its claims are corrected in place below: "the comfort margin beats the gap" holds only at $a_{OV} = -10$, the winner of an exploratory sweep, and card S1.5 showed that on this study the assumed lead braking is the standoff (one lead speed), so that setting is the released -6 with the ego stopping 20.8 m further out; at the ruled -6 the margin scores 0.2230. And " $p = -0.861$ " is a horizon-sum accounting effect, not an inversion of the term. What stands: the difference between strand 1's absolute conflict preference and the released graded one, and the observation that a strand-1 agent needs no braking-margin term. What came of the strand-1 line: card S1.6 (admissibility turns ΔG into the comfort price of the mildest admissible deceleration, and contains a gate as a tolerance on $P(\text{conflict})$), then the gate's derivation in JJ.6e.

2026-09-18, on Jonas's ruling S11.Q3: "Yes, absolutely build it properly." Strand 1 is Engström, Wei, McDonald, Garcia, O'Kelly & Johnson (2024), Resolving uncertainty on the fly: modeling adaptive driving behavior as active inference, *Front. Neurobot.* 18:1341750 — the model docs/waymo_program_revisit.md §1 identifies as the program's own home for the comfort-zone regime, where this project has been using strand 2's collision-avoidance model instead.

Status: a design note. Nothing is authorized and nothing is built. Every specification below is quoted or paraphrased from the paper with its location; every judgment is marked. The rules in §5 become binding only when Jonas has read them.

0 The code situation, because it changes the shape of the work

There is no code release. The paper's data availability statement reads, in full: the original contributions are in the article and supplementary material, and further inquiries should be directed to the corresponding author. No repository, no Zenodo DOI, nothing in the SI beyond the model description.

Two consequences:

- 1 **Ask first.** The corresponding author is Engström, whom Jonas has already spoken to about this work (prompt log, 2026-08-22). If he has the code, this note's §5 becomes a port and the fidelity of a reconstruction stops being something we have to defend in every later claim. Query **S12.Q1**.
- 2 **A reconstruction is feasible anyway**, and §2 says why: the paper specifies the algorithm in prose and gives the preference priors and their defaults in Table 2, and almost all of the machinery already exists in `src/aidriver/`.

[The one repository that does exist in this line, `github.com/ran-wei/interactive_inference` (Wei et al. 2023), is cloned to `external/interactive_inference` and is not this model — it is the learned inverse-RL branch, it needs the *INTERACTION* dataset, and it carries no *LICENSE* file, so it may be read and not copied. `external/README.md` records that.]

1 What strand 1 is, precisely

The objective (paper Eq. 1–2), the same shape as strand 2's:

$$G(\pi) = -E_Q(o|\pi)[\log P(o)] - E_Q(s,o|\pi) D_{KL}[Q(s|o,\pi) \parallel Q(s|\pi)]$$

pragmatic plus epistemic, in one currency, with the action taken from the first step of the policy minimising it.

The state space (Table 1) carries something strand 2 has no analogue of: a **discrete context variable** I — whether a pedestrian is present behind the occlusion — alongside a discrete conflict indicator c , the pedestrian's position and the ego's kinematics. The context is what the epistemic term buys information about.

The preference priors and their defaults (Table 2), quoted:

preference prior	specification	default
speed keeping	Gaussian centred at the speed limit	$\mu = 10 \text{ m/s}$, $\sigma = 1 \text{ m/s}$

preference prior	specification	default
lane keeping	triangular, centred at 0, bounded at the lane boundaries	—
acceleration	Gaussian centred at zero, x and y	$\mu = 0, \sigma = 0.5 \text{ m/s}^2$
conflict	categorical, an absolute preference over "no conflict"	—

The paper adds that these "were set by hand and no systematic model optimization was performed", and that the purpose was to demonstrate the principles rather than to achieve optimal performance.

The implementation: an SIR particle filter with systematic resampling when $N_{\text{eff}} < N/2$; EFE by propagating particles through the transition and observation models; the posterior predictive entropy by **KDE**; policies by **CEM**; a planning horizon of 4 s; a 200 ms step.

The two demonstrations: the occlusion scenario (slow down while uncertain, speed up once the line of sight resolves it, and move laterally to reach the line of sight sooner — an *epistemic action*), and visual time-sharing with a secondary task.

2 What we already have, and what has to be new

piece	where it is	state
kinematic bicycle model, rollout	src/aidriver/bicycle.py	reusable as is
SIR particle filter with ESS resampling and roughening	src/aidriver/agent.py	reusable; strand 1 has no norm conditioning , so the norm branch is switched off rather than replaced
CEM planner with the pedal constraint	agent._cem	reusable; the horizon changes from 30 steps to 4 s
EFE with a pragmatic and an epistemic part	agent.expected_fre_e_energy	reusable in shape; the epistemic term must change (below)
triangular lane-keeping prior	preferences.log_lateral_pref	reusable — strand 1 uses the same form
Gaussian speed and acceleration priors	preferences.log_speed_pref, log_accel_pref	reusable; the σ values change
a categorical conflict preference	—	new: an absolute preference, i.e. an admissibility constraint, not a graded cost
a discrete context variable in the belief	—	new: I, with its own observation model and the line-of-sight geometry
KDE posterior predictive entropy	—	new: strand 2 uses a closed-form Gaussian approximation; strand 1 uses a KDE and the difference is not cosmetic when the belief is multimodal, which is exactly what a context variable makes it

[Opinion: the honest estimate is that the reusable fraction is large but the three new pieces are where the model's behaviour comes from, so "we already have most of it" would be the wrong thing to tell anybody.]

3 The differences from the released model, which are the whole argument

	strand 2 (Nature, collision avoidance)	strand 1 (Frontiers, uncertainty)
σ_v , speed tolerance	0.5 m/s	1.0 m/s — twice as tolerant
σ_a , effort tolerance	0.1 m/s ²	0.5 m/s² — five times as tolerant
collision	−10 000 log units × a severity linear in impact speed	categorical, absolute "no conflict"
braking margin (p_safe)	indicator on $a_{\text{req}} < -a_{\text{max}}$, × −10 000 × severity	does not exist
inverse- τ preference	one-sided Gaussian on τ^{-1} inside p_{coll}	not present
road edge	−15 000	triangular, bounded at the lane boundaries
epistemic value	$\alpha = 1$, closed-form Gaussian approximation	KDE, and it is the point of the model
belief	norm-conditioned particle filter, hand-written per-scenario norms	SIR particle filter, no norm conditioning
context	none	a discrete context variable the epistemic term resolves

	strand 2 (Nature, collision avoidance)	strand 1 (Frontiers, uncertainty)
horizon	30 steps $\times$ 0.2 s = 6 s	4 s

Read the first four rows together and the reformulation note's central proposal is in the authors' own table.

docs/active_inference_reformulation.md §4 item 3 argued that the catastrophe factors belong in the *admissibility of a policy* rather than as a cost ten thousand times the comfort terms, and that the comfort terms need tolerances wide enough to do work. Strand 1 does exactly both. And **strand 1 has no braking-margin term at all** — the term card RE.2 measured at $p = -0.861$ against the share who intervene, and which preferences.py calls "the comfort-zone [2026-09-22: a horizon-sum accounting effect, not a property of the term (review checks §1)] term", does not exist in the model the program puts in the comfort-zone regime. Its role is played by the conflict constraint together with the acceleration prior, which is precisely "can I avoid this without harsh braking", which is card S1.1's comfort margin.

4 What the comfort zone becomes here

In strand 2 the comfort-zone boundary was sought as a **level set of a scalar field** evaluated on a frozen scene. Three cards have now shown that cannot work: RE.2 (no constants), RE.4 (no functionals), and the flat ϵ of RE.1.

In strand 1 the natural object is different: the agent **equilibrates**. Given its preferences and its uncertainty it settles at a speed and a spacing — the paper's own occlusion result is a speed that dips while uncertain and recovers when the uncertainty resolves. So:

The comfort-zone boundary is the operating point a driver's preferences and uncertainty settle at, and a driver's comfort zone is the set of states from which they do not have to leave that operating point.

This is why §2 of docs/waymo_program_revisit.md stops being an obstacle. That section found that our stimuli pin four of the six released preference factors at exactly zero, because the ego never acts. An equilibrium is not evaluated on a frozen scene — it is **simulated**, and a simulated agent does act, so its speed, effort and lane terms come alive. The frozen clip then plays its proper role: a participant saying "I would intervene here" is a datum about where *their* operating point is, not a regression target for a field.

5 The build, as a card ladder with pre-stated rules

Each card can kill the next. Nothing below is authorized.

S1.3 — the agent, and the paper's own Scenario 1. Build the strand-1 preference (the four priors of §1, behind flags defaulting to the released behaviour, standing rule 3), the categorical conflict constraint, the context variable and the KDE epistemic term, in src/strand1/ with property tests in the check() style. Then reproduce the paper's occlusion scenario. **Rule:** the speed trace must show the dip while uncertain and the recovery after the line of sight, and switching the epistemic term off must remove the lateral move toward the line of sight. Those are the paper's two qualitative claims and they are what a reconstruction has to earn.

S1.4 — free following, and this is the cheap one that can stop the line. Card GZ.1 found the *released* configuration does not hold sustained car following: it brakes after 3–5 s with nothing happening. Run the strand-1 agent behind a constant-speed lead for 60 s at several headways. **Rule:** no spontaneous braking below -0.5 m/s^2 in 60 s at a 1.5 s headway, and a settling headway that is finite and repeatable across seeds. If a comfort-zone model cannot sit still behind a lead it cannot have a comfort zone, and the line stops here.

S1.5 — the boundary as an equilibrium. Sweep the driver's own constants — the acceleration tolerance σ_a and the assumed worst case — and read off the settling headway and speed. **Rule:** the settling headway must move monotonically with the effort tolerance, and the range it covers across plausible drivers must contain the headways the studies' participants accept. If the model equilibrates but always at 0.3 s or always at 4 s, it is not measuring this population.

S1.6 — the cut-in. Run the agent through the second study's own stimuli and compare the moment it acts with the moment participants say they would. **Rule:** the project's standing one — held-out wRMSE on the 378 cells, the registered folds and metric, against the gated looming rule's 0.1027 and the gap's 0.1522.

S1.7 — per-driver. Only if S1.6 holds: fit the driver's preference constants per participant, with Wei et al.'s (2023) identifiability treatment adopted rather than rediscovered (docs/waymo_program_revisit.md §4). **Rule:** card TR.1's Spearman interval [+0.407, +0.798] for the per-driver levels across two scenarios.

6 What would falsify this, and one thing that already limits it

Falsification. S1.4 is the cheapest and the most likely to fail: an agent that cannot hold a following distance has no comfort zone. After that, S1.5 — a model that equilibrates but not in the human range is measuring something else. After that, S1.6 on the project's own metric.

The limit that is already known, and it is not about the model. Card S1.2 found that the second cut-in study is the **only** design in this project that varies gap and closing speed independently: study 1's cut-in is one-dimensional (all four axes tried there are rank-identical, Spearman +1.0000), the left turn holds one oncoming speed per cell, the overtake varies lateral clearance. So S1.6 can be run on exactly one design, and any result there cannot be confirmed by transfer within the data we have. That is an argument for a new stimulus set (query WP.Q2) or for the naturalistic data, independent of anything strand 1 does.

7 Queries

- **S13.Q1 (judgment, jonas):** build order. S1.4 is the cheap killer and needs only the preference and the planner, not the context variable or the KDE. Build the minimum for S1.4 first and only then the occlusion scenario for S1.3, or reproduce the paper first so that the reconstruction is validated before anything is claimed from it? *[Opinion: S1.3 first. A result from an unvalidated reconstruction of somebody else's model is worth very little, and S1.3 is the only card here that checks the reconstruction against something the authors published.]*
- **S13.Q2 (judgment, jonas):** the strand-1 defaults were "set by hand" by the authors for a demonstration. Do we take them as given for S1.4 and S1.5, or is the first thing to do a sensitivity over them? They are not calibrated to anything, so treating them as a reference the way the project has treated the Nature constants would repeat the mistake docs/waymo_program_revisit.md §1 describes.
- **S13.Q3 (minor, review):** strand 1 has no norm conditioning at all, while this project has built a cut-in norm (card PN.1, src/comfortzone/norms.py). Does the strand-1 build leave norms out, as the paper does, or is the norm the natural place to put Jonas's "expecting and accepting a closing speed as long as it is far away"? The second is a bigger model than the paper's.
- **S13.Q4 (judgment, review):** the KDE epistemic term is the one piece where a reconstruction can differ from the original silently, because it is an approximation choice inside a term whose scale the released model already gets wrong (docs/method_review.md §6.4). What check would make it trustworthy short of having the authors' code?